Programming - I, Lab 8: Strings and String Operations

1. Read a string and count how many characters in it are **vowels**, **consonants**, **digits**, **spaces** and **other symbols**.

Input	Output

`Hello World 2026!`	`Vowels: 3`
	`Consonants: 7`
	`Digits: 4`
	`Spaces: 2`
	`Others: 1`

2. Read a string and rewrite it so that the characters at even positions (0, 2, 4, ...) are in uppercase and those at odd positions are in lowercase. Characters that are not letters stay unchanged but still occupy a position.

Input	Output
`python`	`PyThOn`
`JKLU jaipur`	`JkLu jAiPuR`

3. JKLU's student portal is rolling out a new sign-up page. Before an account is created, the portal rates the password the student types. It looks for four kinds of characters: an uppercase letter, a lowercase letter, a digit, and a special character (anything that is not a letter, digit or space). A password shorter than 8 characters is always **WEAK**. Otherwise it is **STRONG** if all four kinds are present, **MODERATE** if two or three kinds are present, and **WEAK** if only one kind is present.

   Input: A single line containing the password.

   Output: Print `STRONG`, `MODERATE` or `WEAK`.

   Constraints:
   1 ≤ length of password ≤ 50

Sample Input	Sample Output
`Jklu@2026`	`STRONG`
`jaipur123`	`MODERATE`
`Ab1!`	`WEAK`

4. Read a string and a single character. Print how many times the character occurs in the string and the index of every occurrence. If it does not occur, print `-1` for the positions.

Input	Output
`banana` `a`	`Count: 3` `Positions: 1 3 5`
`banana` `z`	`Count: 0` `Positions: -1`

5. Read a sentence and print the length of each word in it, in order, on one line. Words are separated by one or more spaces, and the sentence may begin or end with spaces.

Input	Output
`  Python   is  fun`	`6 2 3`

6. A Jaipur mall has a parking lane reserved for vehicles registered in Rajasthan. A camera reads the number plate of each car entering the lane. A Rajasthan number plate has exactly 10 characters in the pattern **UUDDUUDDDD**, where U is an uppercase letter and D is a digit, and its first two letters are `RJ` (for example `RJ14CD1234`). Check whether the plate read by the camera is a Rajasthan number plate.

   Input: A single line containing the plate as read by the camera.

   Output: Print `True` if it is a Rajasthan number plate, otherwise print `False`.

   Constraints:
   1 ≤ length of input ≤ 20

Sample Input	Sample Output
`RJ14CD1234`	`True`
`MH12AB0001`	`False`
`RJ14C51234`	`False`

7. Read a string and compare each character from the left with its mirror character from the right. Print the minimum number of characters that must be changed to turn the string into a palindrome.

Input	Output
`abcca`	`Changes needed: 1`
`python`	`Changes needed: 3`
`level`	`Changes needed: 0`

8. Read a string and remove all repeated characters, keeping only the first occurrence of each.

Input	Output
`programming`	`progamin`

9. At the college hackathon, teams receive their problem statement as a coded message. To encode it, each letter of the original message is shifted forward by K positions in the alphabet (with K = 3, a becomes d and c becomes f). If the shift goes beyond z, counting continues again from a (with K = 3, y becomes b). Spaces are left as they are. Write the program that encodes a message so the organisers can prepare the clues.

   Input: The first line contains the message, made up of lowercase letters and spaces only. The second line contains the integer K.

   Output: Print the encoded message.

   Constraints:
   1 ≤ length of message ≤ 100
   1 ≤ K ≤ 25

Sample Input	Sample Output
`attack at dawn` `3`	`dwwdfn dw gdzq`
`xyz abc` `4`	`bcd efg`

10. Read a string and find the character that forms the **longest run** of identical consecutive characters, along with the length of that run. If two runs are equally long, report the one that appears first.

Input	Output
`aabbbbccd`	`b 4`
`xxyyzz`	`x 2`